

Aeration Type Affects Preferential Flow in Golf Putting Greens

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Abstract

Macropore flow has received increased attention in the recent years due to its importance on solute flow through the vadose zone and consequent contamination of groundwater. The effects of man-made macropores resulting from cultivation of turfgrass sites on the mobility of tracer dye (FD&C Blue Dye #1) were studied in free draining greenhouse lysimeters (300 mm I.D. by 463 mm deep) sodded with *Agrostis stolonifera* Huds. ssp *palustris* 'Providence'. Treatments included two soil profiles (305 mm of root zone sand on top of 50 mm coarse sand and 100 mm gravel), simulating a US Golf Association style putting green, and an Arkport sandy loam (Psammentic Hapludalfs coarse loamy mixed), three cultivation practices (shallow hollow tines, 80 mm deep and 19 mm I.D.; deep drill, 220 mm deep and 19 mm I.D.; and high pressure water injection) and two soil moisture regimes (100% and 50% of field capacity). Dye was applied uniformly with a peristaltic pump that delivered water under pressure through a full cone pesticide nozzle. The lysimeters were excavated and the flow pattern of the dye was traced with transparent sheets in various depths. It was observed that in sand at 50% field capacity (FC), dye moved through the aeration holes, even though, the aeration cavities had collapsed at the time of the application. This preferential flow resulted by the hydrophobicity of the thatch layer, which channelled water and dye towards the aeration holes. Preferential flow was also pronounced in the absence of aeration holes. In the sandy profiles that were at 100% field capacity, water and dye movement was more uniform compared to the drier ones. In the sand-50% FC the fastest dye movement was observed in deep drill treatment while in the sand-100% FC the differences between aeration treatments were minor. In sandy loam soil deep tine cultivation resulted in fast movement when the soil was at 100% FC. In contrast, hollow tine cultivation provided reduced dye movement similar to that of the control, while water injection resulted in moderate dye movement. In the sandy loam soil-50% FC all treatments provided fast movement of the dye that had preferential characteristics even in the non-aerated profiles.

INTRODUCTION

In our effort to evaluate the contamination hazard of the turf-applied chemicals such as fertilizers and pesticides, it is essential to accurately describe the pathways of flow. Preferential flow of water and agrochemicals has been established as the main pathway for several agrosystems (Adreini and Steenhuis, 1990) including turfgrasses (Nektarios et al., 1999). Preferential flow is named after the preference of water to flow through specific locations rather than moving uniformly throughout the soil profile (Hill and Parlange, 1972). Preferential flow can occur in coarse textured profiles (Glass et al.,

1989; Nektarios et al., 1999), over inclined layers existing within the profile (funnel flow), in hydrophobic soils (Bauters et al., 1998; Ritsema et al., 1993) and through soil macroporosity. One-dimensional models that are based on the convective-dispersive equation and assume a uniform wetting front have been frequently found to underestimate the travel velocity mainly because of the spatial variability of water flow through the vadose zone.

In turfgrass profiles macropores are in abundance due to the existence of tree root channels and earthworm activity. However, in turfgrass systems and especially in athletic fields, manmade macropores are frequent due to the application of aeration (Nektarios, 1994). Aeration is the cultivation technique where holes are performed with specialized equipment on turfgrass surface in order to alleviate soil compaction and facilitate gas exchange through the soil profile.

The aim of the present study is to visualize the flow of dyed water, which simulates the mobility of water-soluble pesticides: a) through two different profiles (sand and sandy loam); b) using three different aeration techniques; and c) under two soil moisture regimes.

MATERIALS AND METHODS

Sixteen lysimeters were constructed from a PVC pipe (Schedule 80) having a height of 460 mm and internal diameter 300 mm. Eight of the lysimeters were filled with an Arkport sandy loam soil. The other eight lysimeters were filled with 100 mm of gravel at the bottom, 50 mm of coarse sand in the middle and 305 mm of sand root zone simulating a US Golf Association style putting green (USGA Greens Section Staff, 1993).

Compaction was applied to all the lysimeters with a free falling weight technique where a metal weight (5.3 kg) was left to fall on a circular metal plate that fitted tightly on the top of the lysimeters. The weight was dropped 30 times from 550 mm height on the sandy profiles and 10 times from the same height for the Arkport sandy loam profiles. The lysimeters were sodded with washed sod of creeping bentgrass (*Agrostis stolonifera* Huds. ssp *palustris* 'Providence').

Shallow hollow tine cultivation was performed with a tine (95 mm deep and 19 mm I.D.) at a spacing 110 by 110 mm. Deep-drill cultivation holes were made in each lysimeter with a hand drill (375 rpm) and a bit (254 mm length and 19 mm wide) from a commercial unit (Floid-MacCay Deep Drill) at a spacing of 110 by 110 mm. Water injection treatment was performed with a commercial unit (Hydroject 3000, Toro Co., Minneapolis, MN) that resulted in four injections per lysimeter at 76 by 140 mm spacing. Irrigation was applied using a peristaltic pump that delivered water under pressure through a full cone pesticide sprayer nozzle (Sprayco 111820).

Half of the lysimeters from each profile were left to reach field capacity (100% FC) and the other half were left to dry to 50% of field capacity (50% FC). Blue dye (FD&C Blue Dye #1) was then applied at a rate of 25.4 mm·h⁻¹ and left to drain for 24 h. The lysimeters were excavated and the tracer dye was monitored at various depths with the use of transparent sheets.

RESULTS AND DISCUSSION

From the excavation of the lysimeters it was evident that the flow was highly variable and that the moisture content exhibited a significant effect on the mobility of the pesticide tracer. Dye movement in the sandy profiles at 50% FC aerated with hollow tines moved mainly through the aeration holes. However looking at the lower side of the sod, it was evident that the thatch and the sod itself had created a hydrophobic surface, which repelled the dye and channelled it through the aeration holes. It has to be stressed that at the time of the dye application the cavities from all the sand profiles had collapsed but the openings of the aeration holes on the sod surface still existed. Similar effects were observed with the water injection and deep drill cultivation in the sand profiles at 50% FC. Dye reached a depth of 300 mm compared to the 250 mm of the hollow tine cultivation. Again dye moved preferentially through the aeration holes mainly because of

the repellency of the sod. It was noticed that at the bottom of the aeration holes created by the deep drill (200 mm depth), the dye stained only the periphery of the hole but not the hole per se. This could be explained by the sorting of sand particles during the collapse of the aeration cavities where the coarser particles are often the first ones that fall. In such way a lens made of coarse textured particles was created at the bottom of the aeration holes, which under the unsaturated moisture conditions required increased hydraulic head to fill their pores compared with the finer textured materials surrounding the lenses. As a result, dye preferred to move sideways around the bottom of the aeration holes creating this concentric circle effect. The control exhibited increased preferential flow as well as edge flow movement mainly because in this case there were not any aeration holes to channel the dye.

In the sand profiles at 100% FC movement was much more uniform compared with the 50% FC moisture treatment especially in the upper 100 mm. Differences between aeration treatments were not pronounced since the hollow tines reached a depth of 230 mm, deep drill reached a depth of 240 mm and water injection 270 mm. The uncultivated control reached a depth of 300 mm (Table 1).

In the sandy loam soil at 50% FC, dye flow exhibited increased spatial variability through the profile. In the hollow tine aeration treatment, dye moved mainly through the aeration holes and to a lesser degree from the soil matrix. The deepest mobility of the dye was observed at a depth of 320 mm and occurred from one of the aeration holes. In deep drill the aeration holes contributed significantly to the deep movement of the dye. Dye movement through the soil matrix was limited and restricted only to the upper 100 mm of the profile. Similar results were obtained from the hydroject aeration since dye moved mostly from the aeration holes. However, preferential movement of the dye was also observed in the absence of the aeration holes in the untreated profiles and reached a depth of 330 mm. Again the hydrophobic sod surface channelled the dye from specific locations from which most of the flux was guided (Fig. 1).

In the sandy loam soil at 100% FC significant differences were observed between the aeration treatments. In the hollow tine aeration, most of the dye was channelled through the aeration holes but the maximum depth that it was reached was only 170 mm. In contrast, in the deep drill treatment, dye moved preferentially through the aeration holes and reached a depth of 390 mm. In this case soil matrix contributed to the drainage of the dye only in the upper 70-100 mm. Similarly, in water injection, dye moved mainly from the aeration holes and reached a depth of 270 mm. In the untreated profile, dye moved preferentially from the profile and reached a depth of only 170 mm. Edge flow was also observed indicating that even 24 h after irrigation, turfgrass surface had become hydrophobic and exhibited water repellent phenomena (Fig. 1).

CONCLUSIONS

In conclusion, significant spatial variability was observed in dye movement for all treatments including the control, indicating that one-dimensional models are not appropriate for estimating contaminant travel velocity from turfgrass sites. In addition, sampling of turfgrass profiles for pesticide residues must include areas at the bottom and the sides of the aeration holes since in most cases solute movement will take place through them. It is recommended that pesticides should not be applied in sandy loam soils after deep drill cultivation especially in conjunction with increased soil moisture due to the expected fast mobility of the contaminants. In contrast, if pesticides are to be applied on aerated sand profiles the moisture of the profile should be close to field capacity in order to minimize contaminant transport.

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Tables

Table 1. Deepest movement of the dye (mm) as affected by soil type, moisture content and aeration treatment.

	Sand		Sandy Loam	
	50% FC	100% FC	50% FC	100% FC
Hollow Tines	240	220	340	180-370s
Deep Drill	300	240	330	390
Water Injection	270	270	300	270-320w
Control	260	300	330	180-350s

s indicates that side flow was observed

w indicates the existence of a worm hole

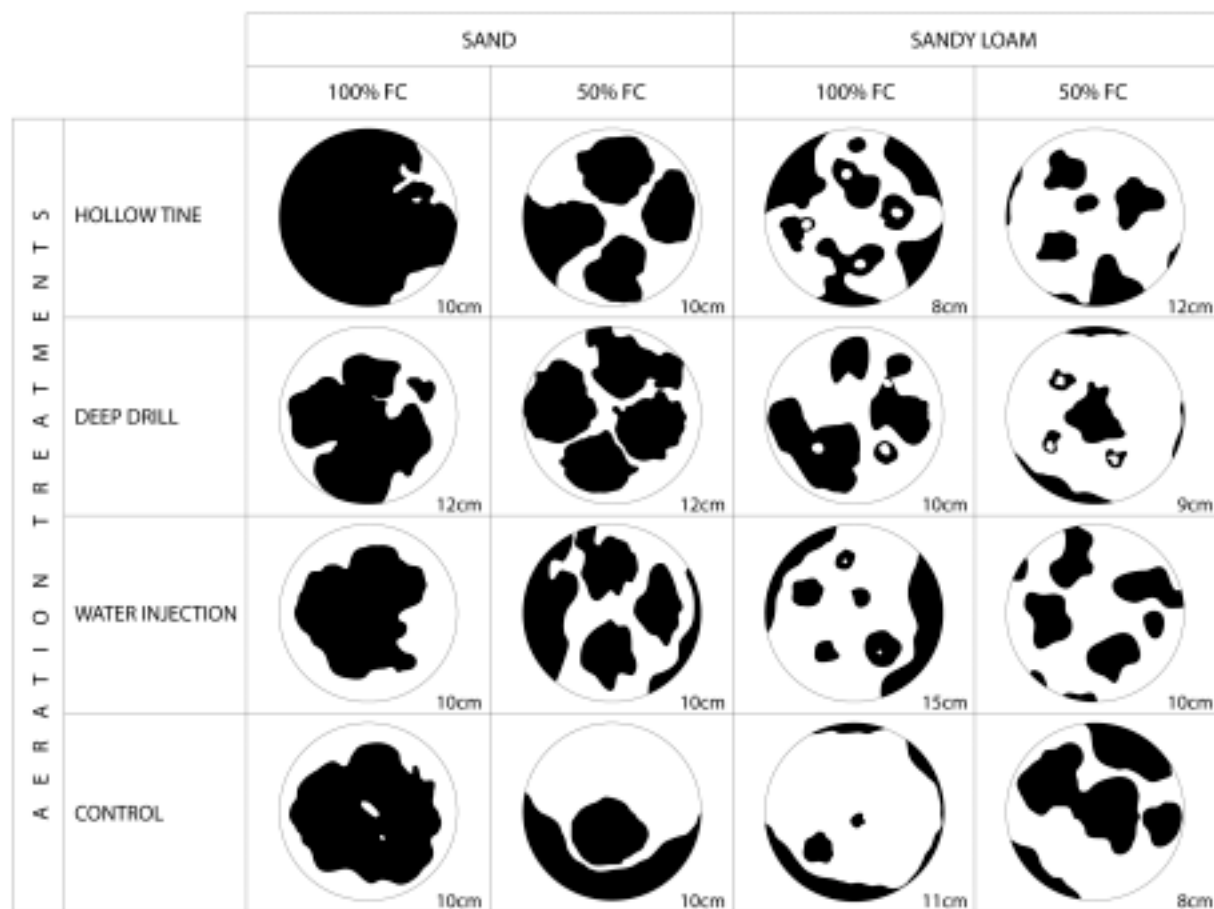


Fig. 1. Dye movement as affected by: a) aeration treatment (hollow tine, deep drill, water injection and uncultivated), b) soil type (sand or sandy loam) and c) by moisture level (100% and 50% of filed capacity). The depth that each dye pattern was measured is defined at the lower right side.